CONTINUAL LEARNING FROM CHIN

THE ONE-LINE SHIFT

From “what to store” to “which weights may change”: 150 of 956 papers now allocate capacity in a chosen subspace instead of replaying data.

LoRA/PEFT 150 · replay 148 · distillation 119
regularisation 94 · architecture 92

中国持续学习研究前沿图谱

What 1,009 papers say the field is actually doing. Replay is no longer the frontier — the unit of research has

1,009
papers matched: “continual learning”, at least one mainland-China author, Jan 2025 – Oct 2026

531
in 2026 alone (Jan–Oct), vs 425 across all of 2025 — the field is accelerating

31.5%
of works are LLM-centric, up from 27.5% in 2025; vision still the largest lane

0
median citations: the frontier is pre-consensus, not settled science

01 THREE SHIFTS

METHOD

Replay lost the frontier.

148 papers still replay. But the 2025–26 method wave is capacity allocation: LoRA/PEFT (150), regularisation (94), architecture (92). The question changed from what to store to which subspace may change.

LoRA · null-space · singular vectors

SETTING

CL moved onto pretrained models.

300 vision-flavoured works, mostly frozen encoders plus task-specific projections, adapters or experts. Exemplar-free is the default; class-incremental toy benchmarks are no longer the main stage.

frozen backbone · adapters · MoE

OBJECT

The object is knowledge itself.

LLM adaptation, agent memory, model merging (24 papers), DP-CL, unlearning. Forgetting is now measured on capabilities — instruction following, reasoning, tool use — not only on labels.

LLM · agents · DP · merging

02 SIX TECHNICAL MOVES

Subspace geometry replaced replay buffers. 01

Null-space projection (O-LoRA lineage) fixes directions, but finite-precision updates still leak into old subspaces. 2026 work turns the step size into a function of accumulated rank — PaLoRA derives s* = sqrt(R/c) — and moves the allocation unit from layers to singular-vector channels (SVC).

Orthogonality is not free. 02

Strict isolation blocks transfer between related tasks; CoDe-LoRA names this the orthogonality dilemma. The fix is partial isolation: adaptive null-space projection plus semantic routing, i.e. shared versus private capacity allocated per task, not per parameter.

Isolation by construction: block-diagonal experts. 03

Monarch-MoE replaces dense LoRA experts with block-diagonal factors joined by a fixed permutation: blockwise gradient isolation at 25–50% of the adapter parameters. CL-MoE and SMoLoRA add task-level and instance-level routing for continual VQA and visual instruction tuning.

Multimodal forgetting is dual. 04

Continual visual instruction tuning degrades visual understanding and instruction following separately (SMoLoRA); CLIP-based CL adds a drifting modality gap (Mind the Gap, ICCV'25). The response is separate routers and explicit gap compensation, not one regulariser.

Agents: memory became the CL substrate. 05

Hippocam consolidates experience along an intent hierarchy; TSRL supervises only failure states and bounds cumulative updates — a rare formal forgetting argument for agent post-training. EmbodiedCLBench supplies the missing evaluation for self-evolving agents.

Generative and private CL are next. 06

Continual diffusion needs gradient projection plus iterative SVD to survive the Markov memory cost; condition-anchored distillation replaces stored prompts with teacher rollouts; DP-SPA shows privatised LoRA training increases forgetting during later adaptation.